A FHIR-to-OMOP Common Data Model ETL Pipeline: Design, Verification, and a Decomposed Concept-Mapping Coverage Metric on the MIMIC-IV-FHIR Demo Dataset

Saileshwar Dumpala
dsailesh2001@gmail.com

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Abstract

Electronic health record systems increasingly expose clinical data through Fast Healthcare Interoperability Resources (FHIR), an interoperability standard designed for point-of-care data exchange, while large-scale observational health research relies on the Observational Medical Outcomes Partnership (OMOP) Common Data Model (CDM), a research-oriented schema designed for standardized, cross-site analysis. This report describes and verifies an open, end-to-end extract-transform-load (ETL) pipeline that maps FHIR clinical resources from the MIMIC-IV Clinical Database Demo on FHIR (100 de-identified patients) into OMOP CDM v5.4, covering the `PERSON`, `VISIT_OCCURRENCE`, `CONDITION_OCCURRENCE`, and `MEASUREMENT` tables, with terminology mapping through the OHDSI standardized vocabularies. I report three contributions beyond the implementation itself. First, I show that a single blended concept-mapping-coverage statistic can misrepresent pipeline correctness, and I present a decomposed metric that separates source-data coverage ceilings from mapping-logic failures, distinguishing a case where low coverage (1.11%) reflects a true source-data limitation, fully verified against two independent data releases, from a case (96.5%) where nearly all records carry a mappable code. A per-patient breakdown shows that the low measurement ceiling is also uneven: 36 of the 100 patients have no mapped measurements, because only emergency-department observations carry LOINC codes in this source. Second, I describe a synthetic-fixture testing methodology developed specifically to satisfy continuous-integration (CI) testing under a real constraint common to OMOP-based projects: OHDSI Athena vocabulary files require individual license acceptance and cannot be redistributed, so CI cannot use real vocabulary content. Third, I document two data-integrity edge cases surfaced by deliberately adversarial fixture design – absent from the real 637-encounter dataset by chance rather than by design – one of which was fixed at the SQL level and one of which is a structural limitation documented with a regression-guarding test rather than silently patched. All quantitative claims in this report were verified against a running instance of the pipeline, not assumed from code review.

1 Introduction

Fast Healthcare Interoperability Resources (FHIR) has become the dominant standard for exchanging clinical data between systems at the point of care [1]. Observational health research, by contrast, is increasingly organized around the OMOP Common Data Model, maintained by the Observational Health Data Sciences and Informatics (OHDSI) collaborative, which standardizes both schema and terminology across participating sites so that analytic code can run unmodified against data collected under different local systems [2]. These two standards serve different pur-

poses – interoperability versus standardized analysis – and an increasing amount of practical data engineering work in health informatics consists of translating from the former into the latter.

This report documents an implementation of that translation for a bounded, verifiable scope: four FHIR resource types (`Patient`, `Encounter`, `Condition`, `Observation`) mapped into four corresponding OMOP CDM v5.4 tables (`PERSON`, `VISIT_OCCURRENCE`, `CONDITION_OCCURRENCE`, `MEASUREMENT`), built and verified against the MIMIC-IV Clinical Database Demo on FHIR, a 100-patient, publicly available de-identified derivative of MIMIC-IV [3, 4, 5]. I treat this as an engineering and methodology report rather than a claim of novel algorithmic contribution: the value of this work is in a fully open, reproducible implementation, an honest accounting of where the pipeline's outputs are and are not trustworthy, and two specific methodological points – coverage-metric decomposition and license-constrained CI design – that generalize to other OMOP ETL efforts built on top of licensed OHDSI vocabularies.

The complete source code, verification queries, and a continuous-integration suite are publicly available; see Section 10.

2 Background

2.1 FHIR and OMOP

FHIR represents clinical information as discrete, linked *resources* (e.g., `Patient`, `Encounter`, `Condition`, `Observation`), each typically serialized as JSON and referencing related resources by identifier. FHIR's design optimizes for point-of-care exchange: a single patient's record, retrieved and updated as care happens.

OMOP CDM, in contrast, is a fixed relational schema (39 tables in v5.4) designed so that identical analytic code – cohort definitions, data characterization, statistical models – can run against any conformant database regardless of the source EHR system. Every clinically meaningful column in OMOP is backed by a *concept*: a row in the `CONCEPT` table drawn from a standardized vocabulary (SNOMED CT for conditions, LOINC for laboratory and clinical measurements, and others), rather than a free- text or locally-coded value. Resolving a source system's local codes to OMOP standard concepts – *concept mapping* – is therefore a required, non-optional step of any FHIR-to-OMOP pipeline, not an optional enrichment.

2.2 Standard vs. source concepts and `CONCEPT_RELATIONSHIP`

OMOP distinguishes a *source concept* (what the original data said – an ICD-9-CM or ICD-10-CM code, a LOINC code, a locally defined code) from a *standard concept* (OMOP's canonical concept for that clinical fact). Every relevant OMOP table stores both (e.g. `condition_concept_id` alongside `condition_source_concept_id`), so no information is lost in translation. The mechanism that resolves a source concept to a standard concept is a lookup in `CONCEPT_RELATIONSHIP` where `relationship_id = 'Maps to'` – not a formula, but a large, OHDSI-curated crosswalk table distributed as part of the Athena vocabulary release.

2.3 Dataset

The MIMIC-IV Clinical Database Demo on FHIR (v2.1.0) is a 100-patient, publicly accessible, de-identified subset of MIMIC-IV re-expressed as FHIR R4 resources [3, 4], itself derived from the base MIMIC-IV Clinical Database Demo [5], both distributed via PhysioNet [6]. This pipeline

consumes the FHIR release directly; the base (non-FHIR) demo was used only as an independent cross-check for one data-coverage claim (Section 5).

3 System Design

3.1 Schema separation

The pipeline uses two PostgreSQL schemas: `staging`, which mirrors the source FHIR resource shape (one JSONB column per resource type, raw and disposable), and `cdm`, the standard OMOP v5.4 relational schema. Keeping these separate means the staging layer can be dropped and reloaded without affecting anything downstream, and no analytic tool is ever pointed at unvalidated, source-shaped data.

3.2 Identity resolution

FHIR resource identifiers are string UUIDs; every OMOP primary key is an integer surrogate. A small set of crosswalk tables (e.g. `staging.person_id_map`, `staging.visit_id_map`), each a `GENERATED ALWAYS AS IDENTITY` table populated with `ON CONFLICT DO NOTHING` semantics, assigns a stable integer to each UUID exactly once. This makes every load in the pipeline idempotent: re- running any stage against a database that already contains data neither duplicates records nor reassigns identifiers.

3.3 Scope

Four FHIR resources map to four OMOP tables (Table 1). `Location`, `Organization`, and `Provider` were explicitly out of scope; `location_id`, `provider_id`, and `care_site_id` remain `NULL` throughout. Medication resources (`DRUG_EXPOSURE`) were deferred as future work (Section 9).

Table 1: Pipeline scope: FHIR resource to OMOP CDM table.

FHIR Resource	OMOP CDM Table
Patient	`PERSON`
Encounter	`VISIT_OCCURRENCE`
Condition	`CONDITION_OCCURRENCE`
Observation	`MEASUREMENT`

4 Implementation

4.1 PERSON and VISIT_OCCURRENCE

`Patient` resources map to `cdm.person`, resolving gender, race, and ethnicity against a small, fixed set of OMOP reference concepts (e.g. concept ID 8507 for male, 8527 for White, 38003564 for Not Hispanic or Latino); where no US Core race/ethnicity extension is present, or a given code (e.g. `ASKU`, "asked but unknown") has no OMOP equivalent, the corresponding concept ID is left at `0` rather than guessed. `Encounter` resources map to `cdm.visit_occurrence`; encounters missing `period.end` are excluded, since `visit_end_date` is `NOT NULL` in the OMOP DDL and no valid value can be synthesized.

4.2 CONDITION_OCCURRENCE and MEASUREMENT

`Condition` resources have no date field of their own in this dataset; `condition_start_date` is instead borrowed from the linked visit's `visit_start_date`, a judgment call made explicit in the pipeline's documentation rather than left implicit in the SQL. ICD-9-CM/ICD-10-CM source codes are reformatted to match `cdm.concept.concept_code` conventions (a decimal point inserted after the third character, e.g. `5715` → `571.5`) and resolved to standard SNOMED concepts via `CONCEPT_RELATIONSHIP`.

`Observation` resources map to `cdm.measurement`. `measurement_date` is taken directly from `effectiveDateTime` (unlike `Condition`, `Observation` carries its own date). Value handling covers three shapes present in the data: `valueQuantity` (numeric, with optional `referenceRange`), `valueString`, and `valueCodeableConcept` display text as a fallback. `value_source_value` is truncated to 50 characters via `LEFT(..., 50)` to satisfy the corresponding `varchar(50)` schema constraint – a deliberate, documented lossy step rather than a silent one; the untruncated text is not preserved elsewhere in the pipeline. Observations are resolved to standard concepts only when tagged with a coding whose `system` is `http://loinc.org`; other coding systems present in the source data have no standard-vocabulary target in this pipeline's scope.

5 Concept-Mapping Coverage: A Decomposed Metric

A single concept-mapping coverage percentage conflates two structurally different situations: a source record that simply never carried a mappable code, and a source record whose code exists but has no corresponding entry in `CONCEPT_RELATIONSHIP`. Reporting only a blended figure can misrepresent which of these is occurring, with direct consequences for how a reader should interpret pipeline correctness.

For `CONDITION_OCCURRENCE`, this distinction does not materially matter: staged condition records overwhelmingly carry an ICD-9-CM or ICD-10-CM code, so a single ratio is already representative. Of 5,051 loaded rows, 4,872 (96.5%) resolved to a standard SNOMED concept; the unmapped 179 rows (3.5%) are predominantly administrative or status codes (e.g. "Do not resuscitate status") with no corresponding disease concept, not resolution failures.

For `MEASUREMENT`, the distinction is material. Of 813,540 loaded observations, only 9,042 (1.11%) carry a coding tagged `system = http://loinc.org`; the remaining 98.9% use MIMIC's own local chartevents/labevents item dictionaries. This was confirmed against two independent sources – the FHIR export's coding systems and the base (non-FHIR) MIMIC-IV Clinical Database Demo's `d_labitems`/ `d_items` tables – neither of which provides a LOINC crosswalk for these local dictionaries. Reported alone, 1.11% coverage could plausibly be read as a mapping-logic defect. Decomposing the statistic shows otherwise: of the 9,042 rows that carried a real LOINC code, all 9,042 (100%) resolved successfully via `CONCEPT_RELATIONSHIP`. The mapped-row count equals the LOINC-coded-row count exactly. The 1.11% figure is therefore a source-data coverage ceiling, verified rather than assumed, and not evidence of a defect in the mapping implementation.

I suggest this decomposition – coverage among rows carrying a vocabulary-eligible code, reported separately from overall table coverage – as a general practice for OMOP ETL reporting whenever a source system's code coverage for a given domain and vocabulary is not close to complete, since a single blended number is otherwise ambiguous between two causes with very different engineering implications.

5.1 Distribution across patients

The overall figures above do not show who is affected. I computed the same two ratios for each of the 100 patients with `scripts/per_patient_coverage.py`, which writes `per_patient_coverage.csv` in the repository. The script's totals reproduce the overall figures exactly (9,042 of 813,540 measurements; 4,872 of 5,051 conditions). Table 2 summarizes the per-patient results.

Table 2: Per-patient concept-mapping coverage (100 patients). Coverage is mapped rows divided by source resources for that patient.

	MEASUREMENT	CONDITION_OCCURRENCE
Median per-patient coverage	0.52%	98.1%
Minimum	0%	62.5%
Maximum	26.76%	100%
Patients with no mapped rows	36	0
Patients below 100%	100	57

Measurement coverage is low for almost every patient and zero for more than a third of them. The cause is visible in the source files. Of the nine observation files loaded, only the two emergency-department files (`MimicObservationED`, 2,742 observations, and `MimicObservationVitalSignsED`, 6,300 observations) use LOINC, and together they hold exactly the 9,042 LOINC-coded observations. The other seven files (chart events, lab events, output events, datetime events, and three microbiology files) use MIMIC's local code systems. I checked this patient by patient in the raw FHIR files: 64 patients have an emergency-department encounter, exactly those 64 have LOINC-coded observations, and the 36 patients without such an encounter have none. Because a measurement row can only map through a LOINC code, and every LOINC-coded row resolved, a patient's mapped count equals their LOINC-coded count. The 36 patients with no emergency-department encounter are therefore the 36 with no mapped measurements.

This means measurement coverage here depends on a patient's care pathway, not only on the vocabulary. An analysis restricted to standard measurement concepts would have no measurements at all for these 36 patients, and neither the blended figure (1.11%) nor the decomposed one (100% within the LOINC-coded subset) reveals that. Condition coverage is far more even (median 98.1%, minimum 62.5%), because condition records overwhelmingly carry ICD codes.

6 Testing Methodology

6.1 The licensing constraint

OHDSI Athena vocabulary files require individual license acceptance prior to download and are not licensed for redistribution. This rules out both of the obvious approaches to CI testing: automating a real vocabulary download (no scriptable license acceptance, and no right to redistribute the result even if it could be scripted), and committing any slice of a real `CONCEPT/CONCEPT_RELATIONSHIP` extract to a public repository.

6.2 Synthetic fixture design

The pipeline's CI suite instead constructs a small, entirely synthetic fixture: fabricated FHIR resources (3 patients, 4 encounters, 4 conditions, 7 observations) and a structurally valid but fabri-

cated OMOP vocabulary subset, run through the pipeline's real, unmodified code. The vocabulary fixture distinguishes two categories of concept differently. Fixed reference concepts that pipeline SQL hardcodes by numeric ID (gender, race, ethnicity, visit type, type concept, and concept ID 0) use their real OMOP IDs, since these are stable identifiers referenced directly by the transform logic, not the copyrighted curated content of a vocabulary. Source and standard clinical concepts (ICD-9-CM/ICD-10-CM/SNOMED/LOINC) are fully fabricated, since these are resolved dynamically by code lookup rather than hardcoded, and fabricating them avoids any licensing concern entirely.

The fixture was deliberately constructed to include adversarial cases rather than a happy path alone: one encounter is missing `period.end` (excluding it from `VISIT_OCCURRENCE` by design), and one condition and one observation are linked to that excluded encounter – a combination never exercised by the real 637-encounter dataset, in which zero encounters were missing a date.

6.3 Two edge cases found

Running the adversarial fixture surfaced two distinct behaviors, handled differently based on the OMOP schema's own constraints.

MEASUREMENT (fixed). `visit_occurrence_id` was originally resolved from the visit ID crosswalk alone, which assigns an identifier to every encounter unconditionally, regardless of whether that encounter's visit actually loaded into `cdm.visit_occurrence`. Against the fixture, this produced a foreign-key violation that aborted the entire multi-row `INSERT...SELECT` statement – meaning a single excluded encounter could have zeroed out all 813,540 real measurement rows had one existed in the production data. The fix changes the join against `cdm.visit_occurrence` to a `LEFT JOIN`, selecting `NULL` when the visit never loaded. This is valid because `MEASUREMENT.visit_occurrence_id` is nullable and `MEASUREMENT` carries its own date independent of the visit. The fix was reverified as a no-op against the real Phase 3 load (zero of 637 real encounters were excluded, so the join always resolved).

CONDITION_OCCURRENCE (documented limitation, not fixed). The analogous case for conditions silently drops the row instead of failing: no error, no log entry, simply absent from `cdm.condition_occurrence`. This cannot be fixed the same way, because `condition_start_date` is `NOT NULL` and `CONDITION_OCCURRENCE` has no date field of its own to fall back on – it borrows the visit's date by design (Section 4.2). A `LEFT JOIN` here would only convert a foreign-key failure into a `NOT NULL` failure. This is documented as a genuine structural limitation of the current design, not patched, and guarded by a regression test asserting the exact expected row count so that a future change to this behavior is noticed rather than passing silently.

6.4 CI architecture

Continuous integration (GitHub Actions) provisions PostgreSQL 16 via a native `services:` container, applies the OMOP v5.4 DDL, loads the fixture through the same scripts and SQL used against real data, and asserts 20 exact expected values (row counts, per-patient gender/race/ethnicity resolution, both edge-case behaviors above, concept-mapping counts, truncation, and referential integrity) via a dedicated test script. All environment-specific values (database host, port, credentials, source data directory) are parameterized via standard `PG*` environment variables read natively by `psql`, so local and CI execution share identical code paths.

7 Results

Table 3 summarizes verified pipeline output on the real 100-patient dataset. Every figure was confirmed by direct query against a populated database, not derived from code inspection alone.

Table 3: Verified pipeline output, MIMIC-IV Clinical Database Demo on FHIR.

Table	Rows loaded	Notes
`cdm.person`	100	0 orphaned references
`cdm.visit_occurrence`	637	0 orphaned `person_id`
`cdm.condition_occurrence`	5,051	96.5% concept-mapped
`cdm.measurement`	813,540	1.11% mapped; 100% within the LOINC-coded subset

CI (GitHub Actions) passes against the synthetic fixture on every push, confirmed via an actual GitHub Actions run rather than local simulation alone. One CI-only failure occurred during development: two shell scripts lost their git-tracked executable bit on a Windows/OneDrive-backed checkout, causing an exit-126 permission error when invoked as `./script.sh`. This was fixed at the git-index level (`git update-index --chmod=+x`) and defensively, by invoking scripts via `bash script.sh` rather than relying on the executable bit at all.

7.1 Demonstration query

As a concrete illustration that the loaded CDM supports genuine cross-table analytic questions, I ran a query joining `CONDITION_OCCURRENCE`, `CONCEPT`, `VISIT_OCCURRENCE`, and `MEASUREMENT` to compute, for each standard condition concept, the number of distinct patients diagnosed with it and their average count of measurements recorded during the diagnosing visit. Table 4 shows the top results. The leading chronic-disease conditions (hypertension, hyperlipidemia, type 2 diabetes) are consistent with expectations for an ICU-derived cohort, and average measurement intensity per visit is higher for acuity-associated conditions (acidosis, thrombocytopenia, acute kidney injury) than for chronic-but-stable conditions (hypertension, tobacco dependence), a pattern consistent with clinical plausibility though not independently validated against ground truth.

Table 4: Demonstration query output (top 8 of 20 by patient count).

Condition	Patients	Diagnoses	Avg. meas./visit
History of event	72	320	301.0
Essential hypertension	55	134	186.9
Hyperlipidemia	47	114	267.3
Acute kidney injury	32	77	440.7
Type 2 diabetes mellitus	30	102	224.2
Atrial fibrillation	27	68	363.1
Anemia	26	45	401.2
Urinary tract infectious disease	23	47	318.9

8 Limitations

I state these plainly rather than minimize them. MEASUREMENT concept-mapping coverage (1.11%) is a hard ceiling imposed by the source data, not something further mapping logic can improve without a different data source or a manually constructed local-code-to-LOINC crosswalk, which was out of scope for this work. A condition whose linked encounter is missing a start or end date is silently absent from CONDITION_OCCURRENCE, a behavior never observed in the real 100-patient dataset (zero such cases existed) but present by construction and guarded by a regression test. LOCATION, ORGANIZATION, and PROVIDER are not modeled. CI validates pipeline *code* correctness against known synthetic inputs; it cannot and does not validate real-world concept-mapping coverage or data quality, which were verified separately against the real load. DRUG_EXPOSURE is not implemented. The per-patient analysis covers 100 patients from a single demo export. The pattern that only emergency-department observations carry LOINC codes is a property of how this export assigns code systems, and may not hold for the full MIMIC-IV-on-FHIR release or other sources. The link between the 36 zero-coverage patients and the 36 patients without an emergency-department encounter rests on matching counts and on equal mapped and LOINC-coded totals, not on a row-by-row join of person_id to source patient identifiers.

9 Future Work

Planned extensions include DRUG_EXPOSURE via RxNorm; a comorbidity co-occurrence network over CONDITION_OCCURRENCE (nodes as standard condition concepts, edges weighted by shared-patient count), as a first network-analysis layer over the verified CDM data; a predictive model (e.g. readmission or in-hospital mortality risk) using PERSON/VISIT_OCCURRENCE/CONDITION_OCCURRENCE/MEASUREMENT as features, as a first analytical/ML layer beyond ETL and standards mapping; and extension to a wearable-native source (e.g. WESAD or PPG-DaLiA), mapping physiological signals into FHIR Observation/Device and then into MEASUREMENT.

10 Conclusion

I presented a verified, open FHIR-to-OMOP CDM v5.4 ETL pipeline covering four clinical resource types on the MIMIC-IV-FHIR demo dataset, with every quantitative claim confirmed against a running database rather than assumed. Beyond the implementation, I argue for decomposing concept-mapping coverage metrics to separate source-data ceilings from mapping-logic defects, and I describe a synthetic-fixture CI methodology that tests real pipeline code under the real licensing constraints that apply to any OMOP ETL project built on OHDSI's Athena vocabularies. Two edge cases surfaced by deliberately adversarial test design, absent from the real dataset only by chance, illustrate a broader point: an ETL pipeline validated only against the data it happens to have seen carries untested assumptions that a sufficiently different cohort could violate.

Availability

Source code, SQL, tests, and CI configuration are publicly available at https://github.com/Saileshwar1625/fhir-omop-etl (tag v1.0.2).

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